TypeScript

JavaScript with type checking before running

THE MENTAL MODEL

A blueprint catches mismatched connectors before the device starts.

CODE → TYPE CHECK → JAVASCRIPT → RUN

READ THE CODE

```
const values: number[] = [10,20,30];
function add(xs: number[]): number {
  return xs.reduce((a,b) => a+b, 0);
}
console.log(add(values));
```

FOUR CLUES TO READ IT

: number[] requires an array of numbers.

function add(...) declares a named function.

: number describes the returned value.

Type annotations are erased when converted to JavaScript.

WHAT MAKES IT DIFFERENT

Checks many mistakes before code runs, but does not validate external data at runtime.

Uses the JavaScript runtime; TypeScript is not a faster execution engine.

WHERE IT FITS

Large web applications, shared codebases, editor assistance.

WATCH OUT

A type declaration is not a security check on input received from a network.

TRY ALONE • 60 SEC

Replace one number with "hello". What error appears before running?